

Regularized regression

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Outline: Regularized regression

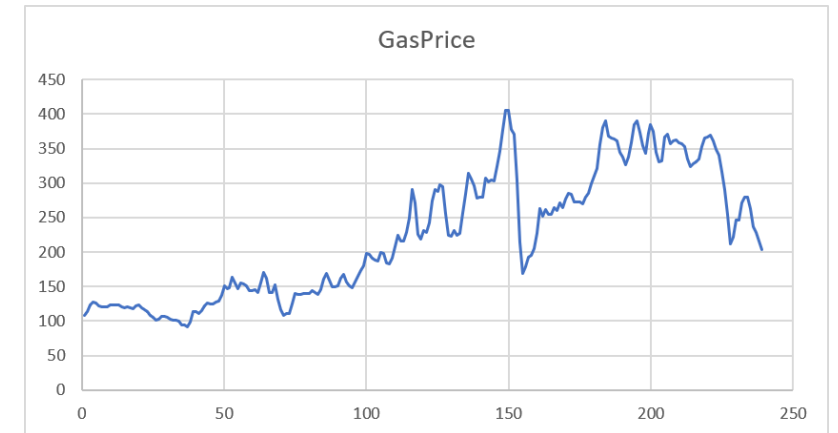
- (25) Motivation
 - Feature scaling
 - OLS limitations
- (30) Ridge regression
- (30) Lasso regression
- (30) Elastic net
- (20) Hyperparameter tuning & Model comparison
 - R demo
- (15) Extensions: Group & Sparse Group Lasso

Date	Year	Month	GasPrice	L1_Crude	L1_GasPrice	L1_SP500	L1_PDI	L1_RR_sales	L1_CPI	L1_Housing	L1_Unemployment	L1_Demand
15-FEB-1996	1996	2	108.9	18.86	109	636.02	5660	134926	154.4	1467	5.6	33.6967
15-MAR-1996	1996	3	113.7	19.09	108.9	640.43	5712.9	136781	154.9	1491	5.5	34.7257
15-APR-1996	1996	4	123.1	21.33	113.7	645.5	5748.8	137527	155.7	1424	5.5	36.6857

Sheet: data

Example: Gas Price

- Dependent variable `GasPrice` (in cents per U.S. gallon)
- 9 Predictors. All variables **lagged** by one month
 1. Price of unleaded gasoline, in cents per U.S. gallon
 2. The price of crude oil, in USD per barrel
 3. The SP500 Stock Index, end-of-month close
 4. Personal disposable income, in billions of current dollars
 5. Consumer price index for all urban consumers, indexed at 100 over 1982-1984, not seasonally adjusted
 6. Retail Sales: real retail sales in millions of USD, deflated by CPI
 7. Unemployment rate, not seasonally adjusted
 8. Housing starts, seasonally adjusted annual rate
 9. Demand for gasoline, in thousands of barrels per day.



Example: Gas Price

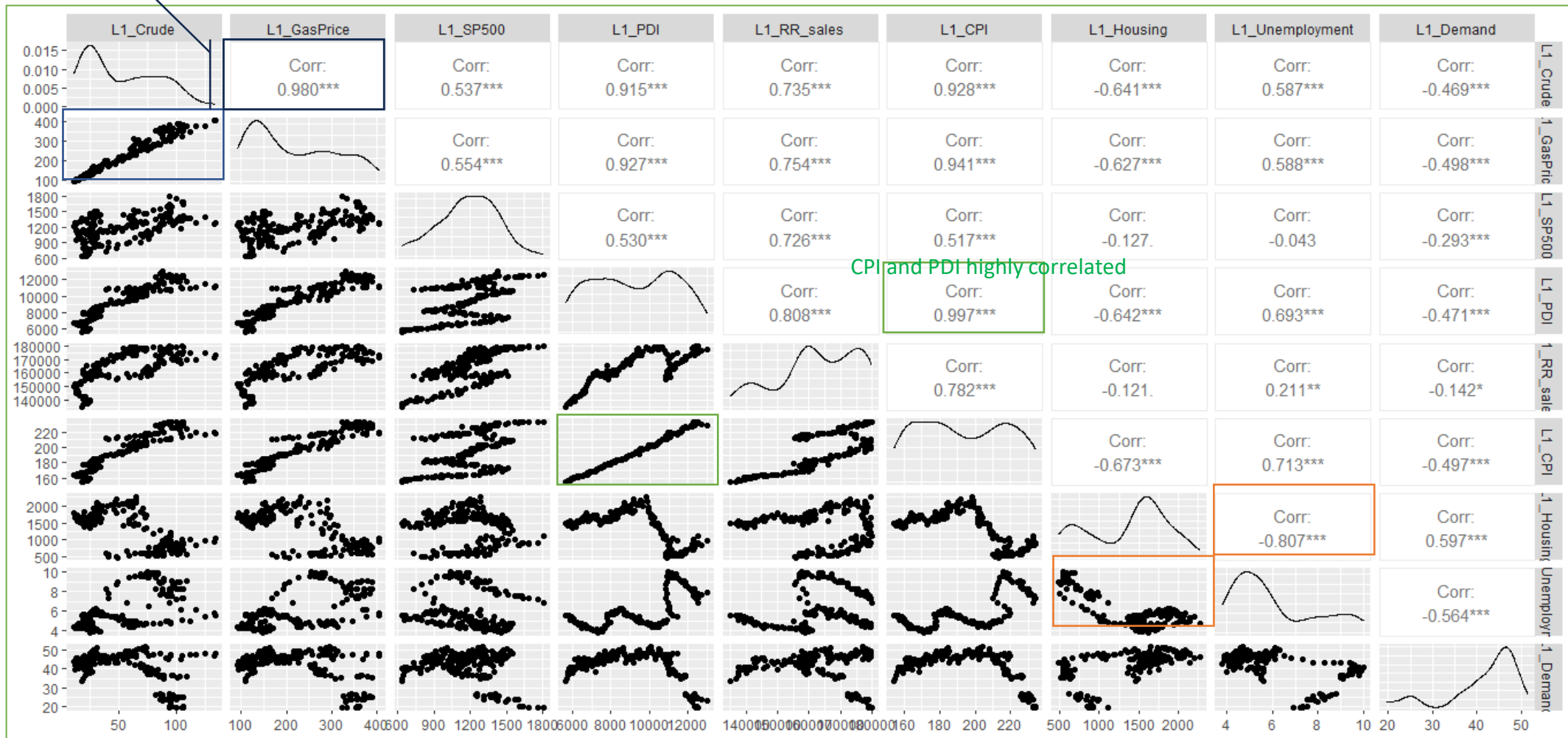
- Dependent variable `GasPrice`
 - Let y_i be the observation in record (row) i
- $k = 9$ predictors, $x_{1,i}, x_{2,i}, \dots, x_{k,i}$
- Model selection: $2^9 - 1 = 511$ “all possible subsets” regression models!

				y_i	$x_{1,i}$	$x_{2,i}$	...				$x_{7,i}$	$x_{8,i}$	$x_{9,i}$
Set	Date	Year	Month	GasPrice	L1_Crude	L1_GasPrice	L1_SP500	L1_PDI	L1_RR_sales	L1_CPI	L1_Housing	L1_Unemployment	L1_Demand
Train	15-FEB-1996	1996	2	108.9	18.86	109	636.02	5660	134926	154.4	1467	5.6	33.6967
Train	15-MAR-1996	1996	3	113.7	19.09	108.9	640.43	5712.9	136781	154.9	1491	5.5	34.7257
Train	15-APR-1996	1996	4	123.1	21.33	113.7	645.5	5748.8	137527	155.7	1424	5.5	36.6857
Train	15-NOV-2013	2013	11	324.3	100.34	324.4	1730.34	12487.0	179307	233.340	921	7.2	19.9309
Train	15-DEC-2013	2013	12	327.6	93.86	324.3	1805.81	12539.1	179777	233.069	1104	6.9	19.9618
Test	15-JAN-2014	2014	1	331.30	97.63	327.60	1848.36	12576.3	180071	233.0	1010	6.7	15.7499
Test	15-FEB-2014	2014	2	335.60	94.62	331.30	1782.59	12650.8	178051	233.9	902	6.6	15.1172
Test	15-OCT-2015	2015	10	229.00	45.48	236.50	1920.03	13640.8	188306	237.9	1189	5.1	21.2742
Test	15-NOV-2015	2015	11	215.80	46.22	229.00	2079.36	13688.1	187883	237.8	1073	5.0	21.4641
Test	15-DEC-2015	2015	12	203.80	42.39	215.80	2080.41	13719.9	188187	237.3	1171	5.0	20.7515

Multicollinearity

Lag 1 of GasPrice and Lag 1 of Crude price highly correlated

```
dtr <- d %>%
  filter(Set == 'Train') %>%
  select(-c(Set, Date, Year, Month))
dtr %>% GGally::ggpairs(columns = 2:10)
```



Regularized Regression: Why?

- Problem: Too many correlated predictors → unstable OLS
- Question: Which variables should we keep?
- Solution: Regularized regression shrinks or removes weak predictors by adding penalty associated with large coefficients.
 - **Ridge**: shrinks coefficients but generally retains all predictors.
 - **Lasso**: shrinks some coefficients exactly to zero, performing variable selection.
 - **Elastic Net**: combines Ridge-type shrinkage and Lasso-type variable selection.

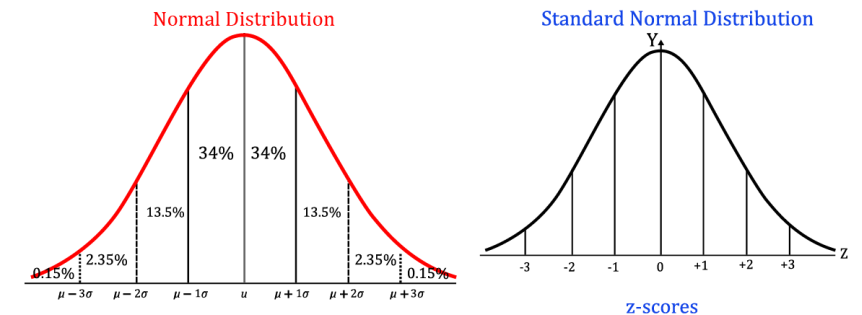
Feature scaling

Standardized before regularization

- Original value of predictor k for observation i : $\tilde{x}_{k,i}$
- Sample mean for predictor k : $\bar{x}_k$
- Sample standard deviation for predictor k : s_k
- Standardized predictor

$$x_{k,i} = \frac{\tilde{x}_{k,i} - \bar{x}_k}{s_k}$$

- After standardization:
 - Each predictor has mean 0 and standard deviation 1.
 - Coefficients are placed on a comparable scale.
 - The penalty does not depend on the original measurement units.
 - Ridge, Lasso, and Elastic Net can shrink predictors more fairly.
- Compute $\bar{x}_k$ and s_k using the training data only, then use the same values to standardized validation and test set.



Ordinary least squares (OLS)

$$\min_{\beta_0, \boldsymbol{\beta}} \text{MSE}(\beta_0, \boldsymbol{\beta}) = \frac{1}{n} \sum_{i=1}^n \left(y_i - (\beta_0 + \beta_1 x_{1,i} + \dots + \beta_k x_{k,i}) \right)^2$$

- Example: Gas Price (Con.)
- Split the data into training and test sets.
- Compute predictor means and standard deviations from the training data. Standardize both training and test predictors using the training-set statistics.
- Estimate the OLS coefficients using the training set.

Sheet OLS

- **Demo:** Use Excel Solver to minimize $\text{MSE}(\beta_0, \boldsymbol{\beta})$ where $\boldsymbol{\beta} = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_9 \end{bmatrix}$

$$\min_{\beta_0, \boldsymbol{\beta}} \text{MSE}(\beta_0, \boldsymbol{\beta}) = \frac{1}{n} \sum_{i=1}^n \left(y_i - (\beta_0 + \beta_1 x_{1,i} + \dots + \beta_9 x_{k,i}) \right)^2$$

- Because the predictors are centered, the intercept, $\hat{\beta}_0 = 216.54$, is the mean of `GasPrice` in the training set.
- Generate fitted values and test forecasts. Compute training and test MSE & RMSE.

Intercept	L1_Crude_z	L1_GasPrice_z	L1_SP500_z	L1_PDI_z	L1_RR_sales_z	L1_CPI_z	L1_Housing_z	L1_Unemployment_z	L1_Demand_z
$\hat{\beta}_0 = 216.536$	24.369	69.886	1.108	50.163	13.685	$\hat{\beta}_6 = -74.153$	-5.158	9.192	-2.356

Concerns with OLS

- Strong correlations among predictors can cause multicollinearity.
- Some coefficient signs are difficult to interpret.
 - Example: the estimated coefficient of CPI is negative.
- Coefficient estimates may be sensitive to small changes in the data.
 - Adding or removing a few observations may substantially change coefficient values.
 - Some coefficients may even change signs.
- Regularization can reduce coefficient sensitivity by shrinking the coefficient estimates.

Intercept	L1_Crude_z	L1_GasPrice_z	L1_SP500_z	L1_PDI_z	L1_RR_sales_z	L1_CPI_z	L1_Housing_z	L1_Unemployment_z	L1_Demand_z
$\hat{\beta}_0 = 216.536$	24.369	69.886	1.108	50.163	13.685	$\hat{\beta}_6 = -74.153$	-5.158	9.192	-2.356

Ridge regression

Ridge regression

- Denote the coefficient as $(\beta_0, \beta_1, \dots, \beta_k) = (\beta_0, \boldsymbol{\beta})$
- Let $x_{k,i}$ be the standardized predictor k in observation i .
- [least squares estimation] Solve the following (unconstrained) optimization problem

$$\min_{\beta_0, \boldsymbol{\beta}} \text{MSE}(\beta_0, \boldsymbol{\beta}) = \frac{1}{n} \sum_{i=1}^n \left(y_i - (\beta_0 + \beta_1 x_{1,i} + \dots + \beta_k x_{k,i}) \right)^2$$

- **Ridge regression (Tikhonov regularization)**

$$\min_{\beta_0, \boldsymbol{\beta}} \text{MSE}(\beta_0, \boldsymbol{\beta}) + \lambda \sum_{j=1}^k \beta_j^2$$

The intercept β_0 is not penalized.

Example: Ridge regression in Excel

				Mean	52.77563	215.5190698	1179.35	9200.1	163077.186	193.625	1362.172093	6.023255814	41.87727488
				Stdev	31.25108	92.49411378	241.8477	2112.82	12302.82724	24.4448	497.3544151	1.801380209	7.830799764
Set	Date	Year	Month	GasPrice	L1_Crude	L1_GasPrice	L1_SP500	L1_PDI	L1_RR_sales	L1_CPI	L1_Housing	L1_Unemployment	L1_Demand
Train	15-FEB-1996	1996	2	108.9	18.86	109	636.02	5660	134926	154.4	1467	5.6	33.6967
Train	15-MAR-1996	1996	3	113.7	19.09	108.9	640.43	5712.9	136781	154.9	1491	5.5	34.7257
Train	15-APR-1996	1996	4	123.1	21.33	113.7	645.5	5748.8	137527	155.7	1424	5.5	36.6857

Feature scaling

coef from OLS	216.536	24.369	69.886	1.108	50.163	13.685	-74.153	-5.158	9.192	-2.356	
coef	216.530	9.080	9.142	4.811	7.918	6.743	8.053	-5.041	4.563	-3.696	
	Ones	L1_Crude_z	L1_GasPrice_z	L1_SP500_z	L1_PDI_z	L1_RR_sales_z	L1_CPI_z	L1_Housing_z	L1_Unemploy	L1_Demand_z	Fitted
	1	-1.08526274	-1.15163079	-2.24657912	-1.67553564	-2.28818836	-1.60465783	0.21077104	-0.23496195	-1.04466659	145.446204
	1	-1.07790299	-1.15271194	-2.2283445	-1.65049796	-2.13741001	-1.58420356	0.25902637	-0.29047494	-0.913262387	145.988295
	1	-1.00622546	-1.10081675	-2.20738089	-1.63350641	-2.07677354	-1.55147673	0.12431358	-0.29047494	-0.662968667	147.775369
	1	-0.93678785	-0.99918866	-2.17153188	-1.64112658	-2.07864303	-1.5269316	0.30929233	-0.23496195	-0.41153841	148.023689

Obj	8473.233242
MSE	8426.769035
Sum of squared coef	0.046464207
lambda	1000.00000

Use Excel Solver to minimize the penalized objective.

Optimization

$$\min_{\beta_0, \beta} \text{MSE}(\beta_0, \beta) + \lambda \sum_{j=1}^k \beta_j^2$$

	Intercept	L1_Crude_z	L1_GasPrice_z	L1_SP500_z	L1_PDI_z	L1_RR_sales_z	L1_CPI_z	L1_Housing_z	L1_Unemployment_z	L1_Demand_z
OLS	$\hat{\beta}_0 = 216.536$	24.369	69.886	1.108	50.163	13.685	$\hat{\beta}_6 = -74.153$	-5.158	9.192	-2.356
Ridge	216.536	0.090	0.090	0.051	0.085	0.070	0.086	-0.057	0.054	-0.045

Ridge regression

- $\min_{\beta_0, \beta} \text{MSE}(\beta_0, \beta) + \lambda \sum_{j=1}^k \beta_j^2$
- Standardize predictors before fitting Ridge.
- Do **not** penalize the intercept β_0 .
- As λ increases, the overall coefficient magnitude decreases and the coefficients are increasingly shrunk toward zero.
- Ridge generally does not set coefficients exactly equal to zero.
- When $\lambda = 0$, Ridge reduces to OLS.

lambda	Sum of squared coef	L1_Crude_z	L1_GasPrice_z	L1_SP500_z	L1_PDI_z	L1_RR_sales	L1_CPI_z	L1_Housing_z	L1_Unempl_z	L1_Demand_z
0 (OLS)	13800.14	24.370	69.886	1.108	50.171	13.685	-74.162	-5.159	9.192	-2.356
10	185.42	5.822	5.894	3.253	5.311	4.479	5.369	-3.422	3.209	-2.666
1,000	0.05	0.089	0.090	0.051	0.085	0.070	0.086	-0.057	0.054	-0.045
1,000,000	0.00	4.05E-05	3.78E-05	1.55E-04	3.06E-05	4.79E-05	6.11E-05	-4.97E-05	-7.76E-05	-5.49E-05

Ridge regression

In-class exercise

1. Add a trend column, $x_{10,i} = i$ where $i = 1, 2, \dots, n$. (Recall that n is the number of observations.)
2. Standardize the new predictor.
3. Fit an OLS regression. Obtain the coefficients.
4. Fit a ridge regression with $\lambda = 10$. Obtain the coefficients. (For now, $\lambda = 10$ is chosen arbitrarily.)

$$\min_{\beta_0, \boldsymbol{\beta}} \text{MSE}(\beta_0, \boldsymbol{\beta}) + \lambda \sum_{j=1}^k \beta_j^2$$

- This example illustrates how the ridge penalty shrinks the estimated coefficients toward zero.
- For forecasting, adding a trend predictor may or may not improve out-of-sample accuracy.

Answer	β_0	$\boldsymbol{\beta}$										Penalty term
	Intercept	L1_Crude_z	L1_GasPrice_z	L1_SP500_z	L1_PDI_z	L1_RR_sales_z	L1_CPI_z	L1_Housing_z	L1_Unemployment_z	L1_Demand_z	Trend_z	$\sum_{j=1}^{10} \beta_j^2$
Coef. from OLS	216.54	23.82	66.64	1.85	53.97	15.19	-54.95	-4.45	11.37	-2.81	-22.41	11834.1
Coef. from Ridge	216.54	5.56	5.62	3.08	4.99	4.23	5.07	-3.24	2.98	-2.54	4.87	190.0

Lasso regression

Regularized Regression

- Denote the coefficient as $(\beta_0, \beta_1, \dots, \beta_k) = (\beta_0, \boldsymbol{\beta})$ The intercept β_0 is not penalized.
- Let $x_{k,i}$ be the standardized predictor k in observation i .
- [least squares estimation] Solve the following (unconstrained) optimization problem

$$\min_{\beta_0, \boldsymbol{\beta}} \text{MSE}(\beta_0, \boldsymbol{\beta}) = \frac{1}{n} \sum_{i=1}^n \left(y_i - (\beta_0 + \beta_1 x_{1,i} + \dots + \beta_k x_{k,i}) \right)^2$$

- **Lasso** (Least Absolute Shrinkage and Selection Operator) regression

$$\min_{\beta_0, \boldsymbol{\beta}} \text{MSE}(\beta_0, \boldsymbol{\beta}) + \lambda \sum_{j=1}^k |\beta_j|$$

Example: Lasso regression in Excel

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Set	Date	Year	Month	GasPrice	L1_Crude	L1_GasPrice	L1_SP500	L1_PDI	L1_RR_sales	L1_CPI	L1_Housing	L1_Unemployment	L1_Demand
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coef	216.530	9.080	9.142	4.811	7.918	6.743	8.053	-5.041	4.563	-3.696	
	Ones	L1_Crude_z	L1_GasPrice_z	L1_SP500_z	L1_PDI_z	L1_RR_sales_z	L1_CPI_z	L1_Housing_z	L1_Unemploy_z	L1_Demand_z	Fitted
	1	-1.08526274	-1.15163079	-2.24657912	-1.67553564	-2.28818836	-1.60465783	0.21077104	-0.23496195	-1.04466659	145.446204
	1	-1.07790299	-1.15271194	-2.2283445	-1.65049796	-2.13741001	-1.58420356	0.25902637	-0.29047494	-0.913262387	145.988295
	1	-1.00622546	-1.10081675	-2.20738089	-1.63350641	-2.07677354	-1.55147673	0.12431358	-0.29047494	-0.662968667	147.775369
	1	-0.93678785	-0.99918866	-2.17153188	-1.64112658	-2.07864303	-1.5269316	0.30929233	-0.23496195	-0.41153841	148.023689

Optimization

$$\min_{\beta_0, \beta} \text{MSE}(\beta_0, \beta) + \lambda \sum_{j=1}^k |\beta_j|$$

Linearizing the absolute values (from optimization class*)

$$\min_{\beta_0, \beta, u} \text{MSE}(\beta_0, \beta) + \lambda \sum_{j=1}^k u_j$$

subject to

$$u_j \geq \beta_j$$

$$u_j \geq -\beta_j$$

$$u_j \geq 0 \text{ (optional; implied by the two above)}$$

Demo: Fill in the rest of the table. Note how coefficients become zero.

lambda	L1_Crude_z	L1_GasPrice_z	L1_SP500_z	L1_PDI_z	L1_RR_sales_z	L1_CPI_z	L1_Housing_z	L1_Unemploy_z	L1_Demand_z
0	24.365	64.596	0.406	0.000	19.176	-23.770	-6.254	9.037	-2.462
0.1	24.522	62.848	0.813	0.000	14.837	-16.311	-4.123	7.418	-2.183
1	24.417	59.480	1.187	0.000	5.483	0.000	0.000	3.111	-1.460
10	21.674	64.047	0.000	0.000	1.260	0.000	0.000	0.000	0.000
100	0.000	41.028	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Lasso regression

In-class exercise (cont.)

Using the standardized predictors from the previous in-class exercise.

Fit a lasso regression with $\lambda = 10$ and record the estimated coefficients.

$$\min_{\beta_0, \beta, u} MSE(\beta_0, \beta) + \lambda \sum_{j=1}^k u_j$$

$$u_j \geq \beta_j, \quad j = 1, 2, \dots, k$$

$$u_j \geq -\beta_j, \quad j = 1, 2, \dots, k$$

- Ridge regression shrinks coefficients toward zero but generally does not set them exactly to zero.
- Lasso regression can set some coefficients exactly to zero, thereby performing variable selection.

Answer	β_0	β									
	Intercept	L1_Crude_z	L1_GasPrice_z	L1_SP500_z	L1_PDI_z	L1_RR_sales_z	L1_CPI_z	L1_Housing_z	L1_Unemployment_z	L1_Demand_z	Trend_z
Coef. from OLS	216.54	23.82	66.64	1.85	53.97	15.19	-54.95	-4.45	11.37	-2.81	-22.41
Coef. from Ridge	216.54	5.56	5.62	3.08	4.99	4.23	5.07	-3.24	2.98	-2.54	4.87
Coef. From Lasso	216.54	20.94	61.16	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.00

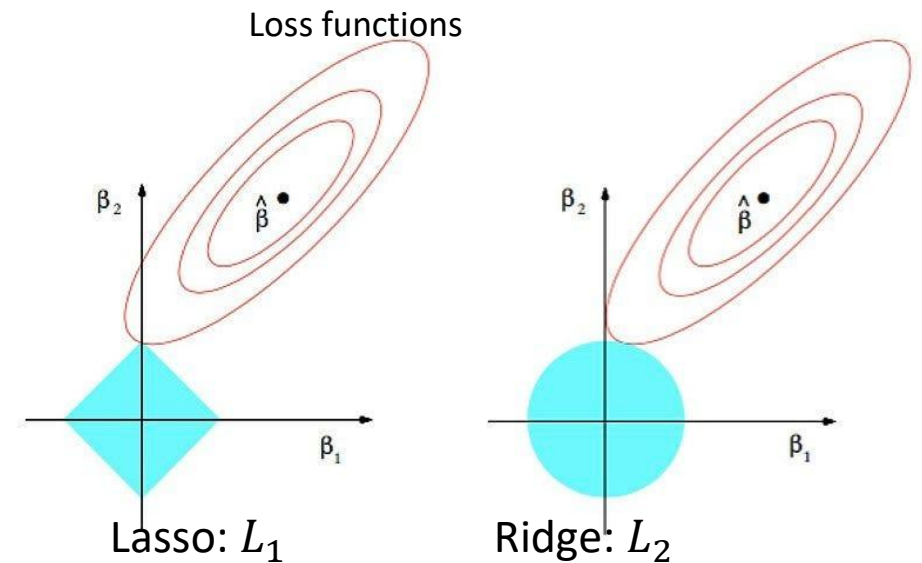
Ridge vs Lasso

- Ridge regression: The penalty is the L2 norm

$$\min_{\beta_0, \beta} \text{MSE}(\beta_0, \beta) + \lambda \|\beta\|_2$$
$$\|\beta\|_2 \leq t$$

- Lasso regression: The penalty is the L1 norm

$$\min_{\beta_0, \beta} \text{MSE}(\beta_0, \beta) + \lambda \|\beta\|_1$$
$$\|\beta\|_1 \leq t$$



- The smooth L_2 boundary usually produces small but nonzero coefficients.
- The corners of the L_1 boundary encourage some coefficients to be exactly zero, producing variable selection.

Elastic Net

Regularized Regression

- Denote the coefficient as $(\beta_0, \beta_1, \dots, \beta_k) = (\beta_0, \boldsymbol{\beta})$
- Let $x_{k,i}$ be the standardized predictor k in observation i .
- [least squares estimation] Solve the following (unconstrained) optimization problem

The intercept β_0 is not penalized.

$$\min_{\beta_0, \boldsymbol{\beta}} \text{MSE}(\beta_0, \boldsymbol{\beta}) = \frac{1}{n} \sum_{i=1}^n \left(y_i - (\beta_0 + \beta_1 x_{1,i} + \dots + \beta_k x_{k,i}) \right)^2$$

- **Ridge regression** (Tikhonov regularization)

$$\min_{\beta_0, \boldsymbol{\beta}} \text{MSE}(\beta_0, \boldsymbol{\beta}) + \lambda \frac{1}{2} \sum_{j=1}^k \beta_j^2$$

- **Lasso** (Least Absolute Shrinkage and Selection Operator) regression

$$\min_{\beta_0, \boldsymbol{\beta}} \text{MSE}(\beta_0, \boldsymbol{\beta}) + \lambda \sum_{j=1}^k |\beta_j|$$

- **Elastic net**

$$\min_{\beta_0, \boldsymbol{\beta}} \text{MSE}(\beta_0, \boldsymbol{\beta}) + \lambda \left[(1 - \alpha) \frac{1}{2} \sum_{j=1}^k \beta_j^2 + \alpha \sum_{j=1}^k |\beta_j| \right]$$

If $\alpha = 1$, Lasso

If $\alpha = 0$, Ridge

where **elastic net mixing parameter** $\alpha \in [0,1]$

Regularized Regression

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- **Ridge regression (Tikhonov regularization)**

$$\min_{\beta_0, \boldsymbol{\beta}} \text{MSE}(\beta_0, \boldsymbol{\beta}) + \lambda \frac{1}{2} \sum_{j=1}^k \beta_j^2$$

- **Lasso (Least Absolute Shrinkage and Selection Operator) regression**

$$\min_{\beta_0, \boldsymbol{\beta}} \text{MSE}(\beta_0, \boldsymbol{\beta}) + \lambda \sum_{j=1}^k |\beta_j|$$

- **Elastic net**

$$\min_{\beta_0, \boldsymbol{\beta}} \text{MSE}(\beta_0, \boldsymbol{\beta}) + \lambda \left[(1 - \alpha) \frac{1}{2} \sum_{j=1}^k \beta_j^2 + \alpha \sum_{j=1}^k |\beta_j| \right]$$

If $\alpha = 1$, Lasso

If $\alpha = 0$, Ridge

where **elastic net mixing parameter** $\alpha \in [0,1]$

Technical note*: Mathematical convenience during optimization. When computing the derivative (gradient), $\frac{\partial \beta_j^2}{\partial \beta_j} = 2\beta_j$. R glmnet and Python scikit-learn include this $\frac{1}{2}$ factor.

Example: Elastic Net in Excel

- As λ increases, the overall coefficient magnitude decreases.
- With $\alpha = 0.5$, the ridge component remains substantial, so at $\lambda = 1000$ the coefficients are extremely small but mostly not exactly zero.
- With $\alpha = 0.75$, the stronger lasso component produces a completely sparse model at $\lambda = 1000$: all slope coefficients are zero.
 - When all slope coefficients are zero, the model predicts only the intercept, which is the training mean of GasPrice because the predictors are centered. *Is this useful for forecasting?*

lambda	alpha	L1_Crude_z	L1_GasPrice_z	L1_SP500_z	L1_PDI_z	L1_RR_sales_z	L1_CPI_z	L1_Housing_z	L1_Unemployment_z	L1_Demand_z
1	0.5	25.828	27.711	7.019	12.230	0.153	13.426	-1.488	3.847	-0.466
10	0.5	11.761	11.896	5.374	9.545	8.234	9.760	-5.530	4.753	-3.951
100	0.5	2.160	2.201	0.805	1.974	1.456	2.010	-0.972	0.880	-0.597
1000	0.5	0.037	0.010	0.001	0.016	0.068	0.009	-0.036	0.103	-0.006
1	0.75	28.217	31.046	2.278	7.892	8.656	9.239	-3.947	0.428	-3.531
10	0.75	15.059	15.296	5.277	10.818	9.190	11.207	-5.607	4.207	-3.641
100	0.75	3.105	3.161	0.489	2.726	1.723	2.776	-0.813	0.625	-0.133
1000	0.75	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

- Increasing λ produces stronger shrinkage.
- Increasing α places more weight on the lasso penalty.

Elastic net

In-class exercise (cont.)

Using the standardized predictors from the previous in-class exercise.

Fit elastic net regressions with $\lambda = 10$ and $\alpha \in \{0.75, 0.99\}$ and record the estimated coefficients.

$$\min_{\beta_0, \beta} \text{MSE}(\beta_0, \beta) + \lambda \left[(1 - \alpha) \frac{1}{2} \sum_{j=1}^k \beta_j^2 + \alpha \sum_{j=1}^k |\beta_j| \right]$$

Equivalent formulation for Excel Solver

$$\min_{\beta_0, \beta, u} \text{MSE}(\beta_0, \beta) + \lambda \left[(1 - \alpha) \frac{1}{2} \sum_{j=1}^k \beta_j^2 + \alpha \sum_{j=1}^k u_j \right]$$

$$u_j \geq \beta_j, \quad j = 1, 2, \dots, k$$

$$u_j \geq -\beta_j, \quad j = 1, 2, \dots, k$$

- A larger mixing parameter α places more weight on the lasso penalty, generally producing a *sparser* model with more coefficients set to zero.
- Note that $\alpha = 0$ gives ridge regression, whereas $\alpha = 1$ gives lasso regression

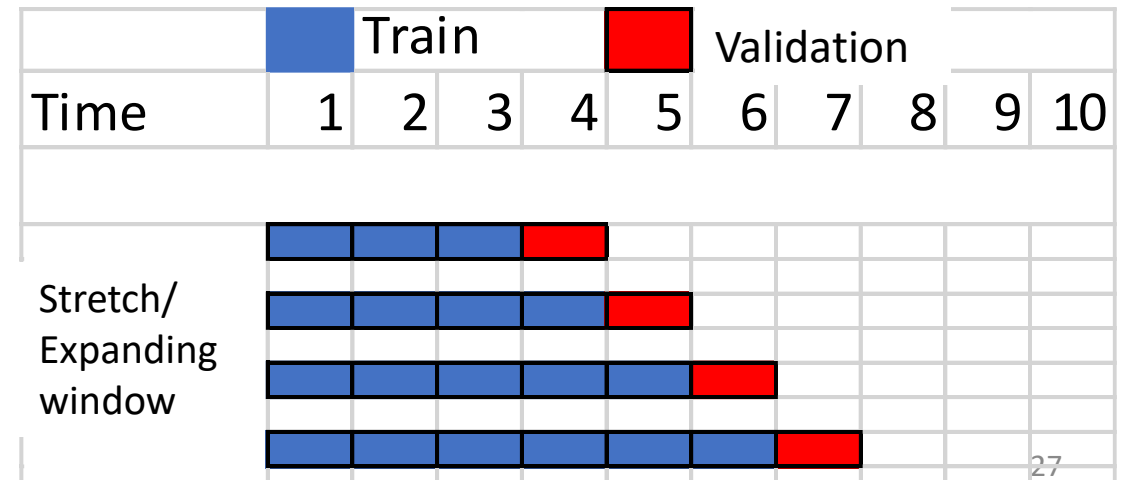
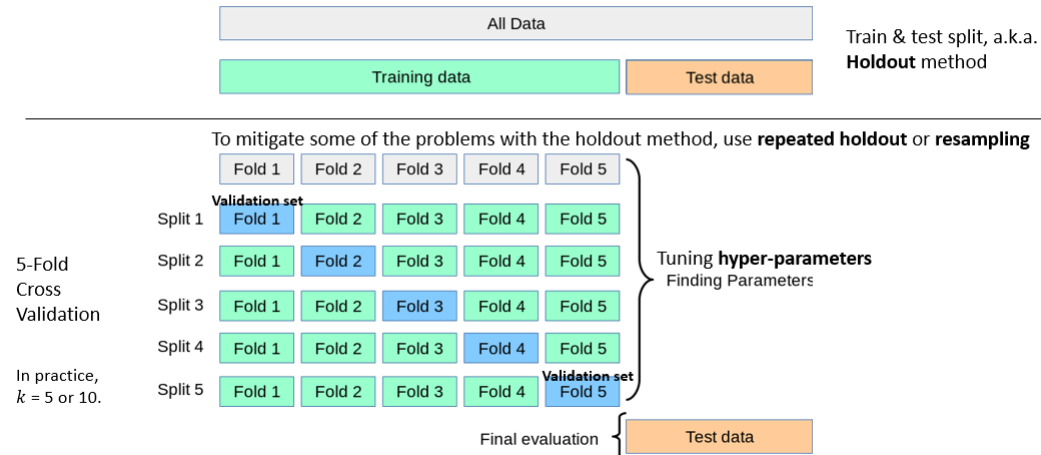
Answer	β_0	β									
	Intercept	L1_Crude_z	L1_GasPrice_z	L1_SP500_z	L1_PDI_z	L1_RR_sales_z	L1_CPI_z	L1_Housing_z	L1_Unemployment_z	L1_Demand_z	Trend_z
Coef. from OLS	216.54	23.82	66.64	1.85	53.97	15.19	-54.95	-4.45	11.37	-2.81	-22.41
Coef. from Ridge	216.54	5.56	5.62	3.08	4.99	4.23	5.07	-3.24	2.98	-2.54	4.87
Coef. from Lasso	216.54	20.94	61.16	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.00
Coef. from Elastic Net $\alpha = 0.75$	216.54	14.53	14.75	5.00	10.03	8.52	10.43	-5.25	3.61	-3.43	5.00
Coef. from Elastic Net $\alpha = 0.99$	216.53	30.37	39.72	0.00	4.50	1.45	5.31	0.00	0.00	0.00	5.00

Hyperparameter tuning

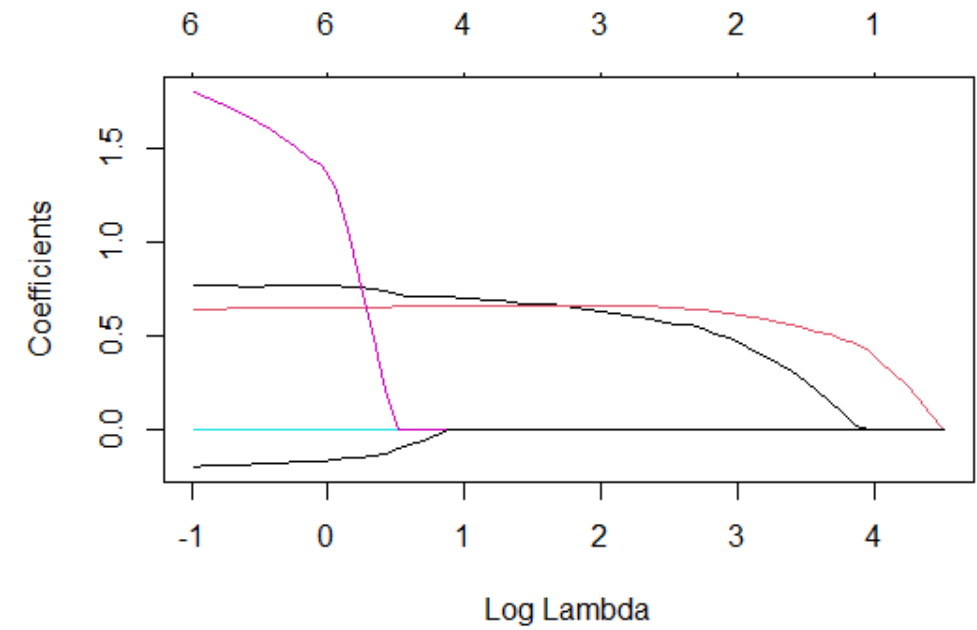
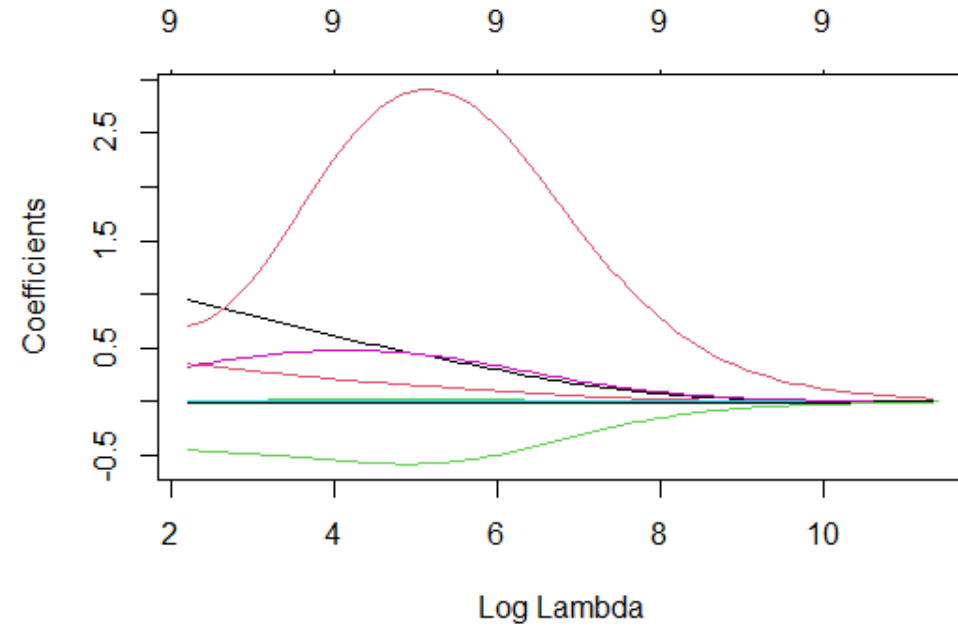
Ridge & Lasso: Hyperparameter tuning λ

- Larger λ means stronger shrinkage.
- Selecting λ by out-of-sample cross-validation (CV)
- Use
 - k-fold CV for approximately independent observations
 - rolling-origin CV for time-series forecasting
- Reserve the test set for final model evaluation.

k -Fold Cross Validation



Ridge vs Lasso



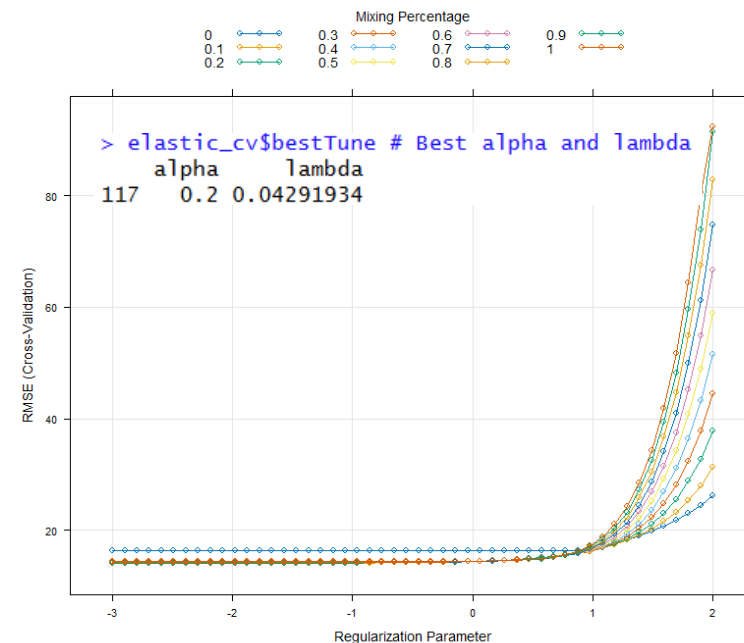
Elastic Net: Hyperparameter tuning λ, α

- $\lambda \geq 0$: Large λ stronger shrinkage
- $\alpha \in [0,1]$: Large α more weight on Lasso penalty
 - $\alpha = 0 \rightarrow$ Ridge
 - $\alpha = 1 \rightarrow$ Lasso
- Tuning approaches
 - Grid search with cross-validation
 - R: `caret` with `method = "glmnet"`
 - Automated hyperparameter optimization
 - Library `Optuna`

Elastic Net: Hyperparameter turning λ, α

R: caret with method = "glmnet"

```
# -----  
# 4. Define 5-fold cross-validation  
# -----  
cv_control <- trainControl(  
  method = "cv",  
  number = 5,  
  verboseIter = FALSE  
)  
  
# -----  
# 5. Define the alpha-lambda tuning grid  
# -----  
tune_grid <- expand.grid(  
  alpha = seq(0, 1, by = 0.1),  
  lambda = 10^seq(-3, 2, length.out = 50)  
)  
  
# -----  
# 6. Tune Elastic Net using training data only  
# -----  
set.seed(123)  
elastic_cv <- train(  
  x = train_model %>% select(all_of(predictors)),  
  y = train_model[[response]],  
  method = "glmnet",  
  trControl = cv_control,  
  tuneGrid = tune_grid,  
  metric = "RMSE"  
)
```



Note

- `glmnet` standardizes the predictors within each CV training fold. The response remains in its original units.

Quick check: How to modify this code?

- Use 10-fold cross validation.
 - Is this resampling scheme appropriate? *Why?*
- Error metric: MAE
- Bigger grid

Train Validation

Time-series cross validation

One-step ahead forecast

1	2	3	4	5	6	7	8	9	10

```
cv_control <- trainControl(  
  method = "timeslice",  
  initialWindow = 3,  
  horizon = 1,  
  fixedWindow = FALSE,  
  skip = 0  
)
```

Expanding window, one-step ahead.
Common for evaluating one-step-ahead forecasts.

Multi-step ahead forecast

1	2	3	4	5	6	7	8	9	10

```
cv_control <- trainControl(  
  method = "timeslice",  
  initialWindow = 3,  
  horizon = 2,  
  fixedWindow = FALSE,  
  skip = 0  
)
```

Expanding window, two-step ahead, overlapping validation sets.
Common for evaluating multi-step-ahead forecasts because it uses every possible forecast origin.

1	2	3	4	5	6	7	8	9	10

```
cv_control_2 <- trainControl(  
  method = "timeslice",  
  initialWindow = 3,  
  horizon = 2,  
  fixedWindow = FALSE,  
  skip = 1,  
  verboseIter = FALSE  
)
```

Expanding window, two-step ahead, non-overlapping validation sets (skip=1).
Uses fewer forecast origins, mainly to reduce computation.

Model comparison

- Elastic Net achieves a lower test RMSE than OLS
- The tuned Elastic Net model retains all predictors but shrinks several coefficient estimates toward zero.
 - Large shrinkage appears for L1_PDI and L1_CPI.
- Regularization does not guarantee that a counterintuitive coefficient sign will change.
 - Elastic Net shrinks the CPI coefficient substantially toward zero, but it remains negative.

	OLS	Elastic Net
RMSE (Test)	18.632	17.742
	OLS Coef	Elastic Net Coef
Intercept	216.536	216.536
L1_Crude_z	24.369	24.964
L1_GasPrice_z	69.886	65.185
L1_SP500_z	1.108	0.982
L1_PDI_z	50.163	22.955
L1_RR_sales_z	13.685	14.017
L1_CPI_z	-74.153	-42.012
L1_Housing_z	-5.158	-4.473
L1_Unemployment_z	9.192	8.059
L1_Demand_z	-2.356	-2.300

Extensions

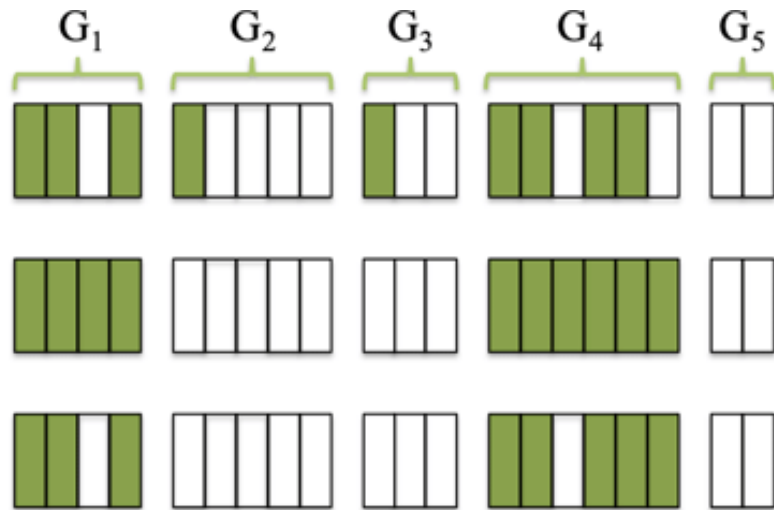
Let v be an m -dimensional vector

$$L_1 \text{ norm: } \|v\|_1 = \sum_{\ell=1}^m |v_\ell|$$

$$L_2 \text{ norm: } \|v\|_2 = \sqrt{\sum_{\ell=1}^m v_\ell^2}$$

Lasso vs Group Lasso vs Sparse Group Lasso

- Let G be the number of groups
- Let β_g be vector of coefficients for group g . Let $\beta = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_G \end{bmatrix}$ be the vector of all coefficients (excluding intercept)
- Let p_g be number of features in that group



Lasso

Group Lasso

Sparse Group Lasso

Lasso penalty

$$\lambda \|\beta\|_1$$

Group penalty

$$\lambda \sum_{g=1}^G \sqrt{p_g} \|\beta_g\|_2$$

$$\alpha \lambda \|\beta\|_1 + (1 - \alpha) \lambda \sum_{g=1}^G \sqrt{p_g} \|\beta_g\|_2$$

Sparse Group Lasso

- Can remove an entire group or individual predictors within a retained group
- Suppose that the predictors are reordered according to G_1, G_2, G_3, G_4 .
- Group 1 Oil and gasoline market $G_1 = \{L1_Crude, L1_GasPrice, L1_Demand\} := \{1,2,3\}$
- Group 2 Consumer purchasing power & spending: $G_2 = \{L1_PDI, L1_CPI, L1_RR_sales\} := \{4,5,6\}$
- Group 3 Broader economic conditions: $G_3 = \{L1_SP500, L1_Housing, L1_Unemployment\} := \{7,8,9\}$

$$\min_{\beta_0, \boldsymbol{\beta}} \{ MSE + \lambda [\alpha \sum_k |\beta_k| + (1 - \alpha) \sum_g \sqrt{|G_g|} \|\boldsymbol{\beta}_{G_g}\|_2] \}$$

$\alpha \in [0,1]$
 $\alpha = 1$: ordinary Lasso
 $\alpha = 0$: Group Lasso
 $0 < \alpha < 1$: Sparse group Lasso

$$\min \text{MSE} + \lambda \alpha \sum_{k=1}^9 |\beta_k| + \lambda(1 - \alpha) [\sqrt{3} \sqrt{\beta_1^2 + \beta_2^2 + \beta_3^2} + \sqrt{3} \sqrt{\beta_4^2 + \beta_5^2 + \beta_6^2} + \sqrt{3} \sqrt{\beta_7^2 + \beta_8^2 + \beta_9^2}]$$

Lasso penalty
Group penalty

where $\text{MSE} = \frac{1}{n} \sum_{t=1}^n (y_t - \beta_0 - \sum_{k=1}^{20} \beta_k x_{k,t})^2$

Group Lasso

- Selects or removes related predictors together as a group.
- Examples
 - Quarterly-seasonality group: Q2, Q3, Q4
 - Categorical predictor: all dummy variables for its levels
 - Holiday baseline no holiday; D1 = major holiday; D2 = minor holiday
 - Fourier pair: $\sin\left(\frac{2\pi kt}{m}\right), \cos\left(\frac{2\pi kt}{m}\right)$
 - Related interaction terms, e.g., $x_4 Q_2, x_4 Q_3, x_4 Q_4$
 - Spline basis functions:
 - Linear spline with knots c_1, c_2 : $x, (x - c_1)^+, (x - c_2)^+$
 - Cubic spline with knots c_1, c_2 : $x, x^2, x^3, (x - c_1)_+^3, (x - c_2)_+^3$

Homework: Sparse Group Lasso (Con.)

- Group 1 Oil and gasoline market $G_1 = \{L1_Crude, L1_GasPrice, L1_Demand\} := \{1, 2, 3\}$
- Group 2 Consumer purchasing power & spending: $G_2 = \{L1_PDI, L1_CPI, L1_RR_sales\} := \{4, 5, 6\}$
- Group 3 Broader economic conditions: $G_3 = \{L1_SP500, L1_Housing, L1_Unemployment\} := \{7, 8, 9\}$
- Group 4 Fourier pairs with $K = 2$, e.g., $\beta_{10}\sin\left(\frac{2\pi t}{12}\right) + \beta_{11}\cos\left(\frac{2\pi t}{12}\right) + \beta_{12}\sin\left(\frac{4\pi t}{12}\right) + \beta_{13}\sin\left(\frac{4\pi t}{12}\right)$
- Group 5 Autoregressive terms $G_5 = \{y_{t-2}, y_{t-3}, y_{t-4}, y_{t-5}, y_{t-6}\} := \{14, 15, 16, 17, 18\}$
- Group 6 Seasonal autoregression terms $G_6 = \{y_{t-12}, y_{t-24}\} := \{19, 20\}$
- Group 7 Moving averages $G_7 = \{MA2(GasPrice), MA12(GasPrice)\} := \{21, 22\}$
- Group 8 Lagged predictor $G_8 = \{L2_SP500, L2_Housing, L2_Unemployment\} := \{23, 24, 25\}$
- Group 9 Trend: $G_9 = \{t, t^2\} := \{26, 27\}$
- Group 10 Average monthly temperature $G_{10} := \{28\}$
- Group 11 Holiday indicator $G_{11} := \{29\}$
- Group 12 Supply disruption indicator $G_{12} := \{30\}$

Write loss function $MSE + \text{Lasso penalty} + \text{Group Penalty}$

- Sparse Group Lasso can be useful when predictors have a meaningful group structure and selection is needed both between groups and within groups.
- Sparse Group Lasso does not automatically enforce hierarchy.